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Project Title	Multi-Channel Marketing Attribution with High-Dimensional Markov Chain Transitions
Week	Week 6

WEEK-6 PROJECT

Task Name:

Multi-Channel Marketing Attribution with High-Dimensional Markov Chain Transitions

Description:

This Week-6 task is essentially a large-scale Markov Chain marketing attribution project. The objective is to process 15 million user path logs, construct a transition probability matrix representing marketing channel interactions, and calculate the Removal Effect of each marketing channel.

1. INPUT: 15 MILLION USER PATH LOGS

Each user journey can contain multiple marketing channels.

Example:

User 101:

Google Ads → Facebook → Email → Direct → Conversion

User 102:

Instagram → Google Ads → Direct → No Conversion

User 103:

Email → Organic Search → Conversion

The dataset should contain at least the following fields:

- user_id
- timestamp

- channel
- conversion

Example:

user_id: 101
channel: Google Ads
timestamp: 10:01
conversion: 0

user_id: 101
channel: Facebook
timestamp: 10:05
conversion: 0

user_id: 101
channel: Email
timestamp: 10:10
conversion: 0

user_id: 101
channel: Conversion
timestamp: 10:15
conversion: 1

user_id: 102
channel: Instagram
timestamp: 11:00
conversion: 0

2. CONSTRUCT USER PATHS

Group all events using user_id and sort them chronologically using the timestamp.

Convert the events into complete user journeys.

Example of a converting journey:

START

↓

Google Ads

↓

Facebook

↓

Email



CONVERSION

Example of a non-converting journey:

START



Instagram



Direct



NULL

Use special states such as:

START

CONVERSION

NULL

3. CALCULATE TRANSITION COUNTS

For every consecutive pair of channels, count how many times users move from one channel to another.

Example:

Google Ads → Facebook: 50,000

Google Ads → Email: 20,000

Google Ads → Direct: 30,000

Create a transition count matrix containing the number of transitions between each pair of channels.

Example:

Google Ads → Facebook = 50,000

Google Ads → Email = 20,000

Google Ads → Direct = 30,000

Facebook → Google Ads = 10,000

Facebook → Email = 25,000

Facebook → Direct = 15,000

Email → Google Ads = 5,000

Email → Facebook = 10,000

Email → Direct = 20,000

4. CONVERT TRANSITION COUNTS INTO PROBABILITIES

Convert the transition counts into transition probabilities.

For every source channel, divide the number of transitions to a destination channel by the total number of transitions from that source channel.

Formula:

$$P(i \rightarrow j) = \text{Number of transitions from } i \text{ to } j / \text{Total transitions from } i$$

Example:

Google Ads has:

Facebook = 50,000

Email = 20,000

Direct = 30,000

Total transitions = 100,000

Therefore:

Google Ads → Facebook = 50%

Google Ads → Email = 20%

Google Ads → Direct = 30%

Each row of the transition probability matrix should approximately add up to 100%.

5. CALCULATE BASELINE CONVERSION PROBABILITY

Calculate the probability that a user starting from START will eventually reach the CONVERSION state.

The Markov Chain consists of:

- Transient states: START and marketing channels
- Absorbing states: CONVERSION and NULL

The transition matrix can be divided into:

Q = transitions between transient states

R = transitions from transient states to absorbing states

Calculate the fundamental matrix:

$$N = (I - Q)^{-1}$$

Then calculate:

$$B = N \times R$$

The conversion probability corresponding to the START state gives the baseline conversion probability.

This value is called:

P_base

6. CALCULATE REMOVAL EFFECT

The Removal Effect is the most important part of the project.

For every marketing channel, remove that channel from the Markov Chain and calculate the new conversion probability.

Example:

Baseline conversion probability:

$$P_{\text{base}} = 0.25$$

Remove Facebook from the Markov Chain.

After removing Facebook:

$$P_{\text{without_Facebook}} = 0.18$$

Calculate the Removal Effect using:

$$\text{Removal Effect} = (P_{\text{base}} - P_{\text{without_channel}}) / P_{\text{base}}$$

For Facebook:

$$\text{Removal Effect} = (0.25 - 0.18) / 0.25$$

Removal Effect = 0.28

Therefore:

Facebook Removal Effect = 28%

A higher Removal Effect means that the channel has a greater contribution to the overall conversion journey.

7. REPEAT FOR EVERY MARKETING CHANNEL

Perform the same calculation for every marketing channel.

Example channels:

- Google Ads
- Facebook
- Instagram
- Email
- YouTube
- Organic Search
- Direct

For each channel:

1. Remove the channel.
2. Rebuild the reduced transition matrix.
3. Recalculate the conversion probability.
4. Calculate the Removal Effect.
5. Store the result.

The example results below are only illustrative:

Google Ads:

Baseline Conversion = 25%

After Removal = 18%

Removal Effect = 28%

Facebook:

Baseline Conversion = 25%

After Removal = 20.5%

Removal Effect = 18%

Instagram:

Baseline Conversion = 25%

After Removal = 23%

Removal Effect = 8%

Email:

Baseline Conversion = 25%

After Removal = 21%

Removal Effect = 16%

YouTube:

Baseline Conversion = 25%

After Removal = 24%

Removal Effect = 4%

Organic Search:

Baseline Conversion = 25%

After Removal = 22.5%

Removal Effect = 10%

These values are examples only and must be calculated using the actual dataset.

8. RANK MARKETING CHANNELS

After calculating the Removal Effect for every channel, rank the channels from highest to lowest.

Example:

1. Google Ads – 28%
2. Facebook – 18%
3. Email – 16%
4. Organic Search – 10%
5. Instagram – 8%
6. YouTube – 4%

The channel with the highest Removal Effect is considered the most influential channel according to the Markov attribution model.

9. RECOMMENDED ARCHITECTURE FOR 15 MILLION RECORDS

Because the dataset contains 15 million records, the project should use an efficient data-processing architecture.

Recommended workflow:

15 Million User Path Logs



Data Cleaning



Sort and Group by User



Construct User Paths



Calculate Transition Counts



Build Transition Matrix



Calculate Transition Probabilities



Calculate Baseline Conversion Probability



Remove Each Marketing Channel



Recalculate Conversion Probability



Calculate Removal Effect



Rank Marketing Channels



Generate Reports and Visualizations

10. RECOMMENDED TECHNOLOGIES

Python can be used as the primary programming language.

Recommended libraries:

- Polars or PyArrow for processing large datasets
- NumPy for numerical calculations
- SciPy for matrix operations
- Pandas for smaller result datasets if required
- Matplotlib for visualization
- Plotly for interactive dashboards if required

For 15 million records, Polars or PyArrow is recommended instead of relying entirely on standard Pandas operations.

11. DATA PROCESSING WORKFLOW

Input:

CSV or Parquet files

Processing:

CSV/Parquet

↓

Polars

↓

Data Cleaning

↓

User Journey Construction

↓

Transition Count Calculation

↓

NumPy/SciPy

↓

Markov Chain Matrix

↓

Baseline Conversion Probability

↓

Channel Removal

↓

Removal Effect Calculation

↓

Channel Attribution Results

12. FINAL PROJECT DELIVERABLES

The completed project should contain:

1. Data preprocessing pipeline.
2. Processing of 15 million user path records.
3. User journey construction.
4. Transition count matrix.
5. Transition probability matrix.
6. Baseline conversion probability.

7. Channel removal calculations.
8. Removal Effect for every marketing channel.
9. Ranked marketing attribution results.
10. Visualization or dashboard.
11. Performance analysis including processing time.
12. Memory usage and scalability analysis.

PROJECT OBJECTIVE

The main objective of this project is to use a high-dimensional Markov Chain model to understand how different marketing channels contribute to customer conversions.

By systematically removing each channel and measuring the resulting decrease in conversion probability, the project identifies the relative importance of each marketing channel.

Final Output:

Marketing Channel

- Transition Probability
- Baseline Conversion Probability
- Conversion Probability After Removal
- Removal Effect
- Attribution Rank